

Two-Level Dual-View Event Relation Graphs for Sentence-Level Media Bias Detection

Anonymous ACL submission

Abstract

This report presents our complete implementation of a two-level dual-view event relation graph architecture for sentence-level media bias detection. Building on Lei and Huang (2024)’s event relation graph approach, we introduce explicit separation of factual events (objectively verifiable occurrences) from interpretive events (subjective characterizations) and process them through hierarchical paragraph-level and document-level graph neural networks. We successfully completed all nine pipeline stages: MAVEN-ERE preprocessing, event extractor training with implicit relation enhancements, event graph construction on BASIL, factual-interpretive classification of 2,500+ events using Gemini API, semantic paragraph detection, implementation of baseline and ablation variants, ten-fold cross-validation training, and comprehensive evaluation. Our full model achieves F1 49.7% on 153 classified BASIL articles, representing +4.0 points improvement over Longformer baseline. The dual-view separation provides the largest contribution (+2.7 F1), followed by hierarchical processing (+1.8 F1) and cross-view attention (+1.2 F1). We match state-of-the-art recall (54.0% vs 54.1%) while identifying controversy framing as a key bias pattern our architecture successfully detects.

1 Project Idea and Motivation

1.1 Motivation

Media outlets increasingly exhibit partisan polarization, making it critical to detect ideological bias at a fine-grained level. While most prior work detects bias at the article or source level, bias often manifests in specific sentences that appear neutral but subtly frame facts to influence readers. The challenge is that bias sentences frequently seem factual when examined in isolation, requiring broader contextual understanding to reveal their persuasive intent.

Consider this example: “*The Senate voted 52-48 to pass healthcare reform. Critics immediately warned the controversial bill could devastate millions.*” The second sentence appears to merely report expert analysis, but its juxtaposition with the factual vote creates negative framing through three mechanisms: (1) temporal adjacency suggesting immediate consequences, (2) implicit causal connection between passage and warnings, and (3) selective presentation of only critical perspectives.

A key insight is that events in bias sentences must be understood in association with other events in the document. Current approaches process all events uniformly, but we observe that the relationship between factual events (objectively verifiable: “voted”, “signed”, “increased 10%”) and interpretive characterizations (subjective: “warned”, “slammed”, “could devastate”) is particularly revealing of bias.

1.2 Project Objectives

This project aimed to improve sentence-level media bias detection by:

Primary Objective: Develop a two-level dual-view event relation graph that explicitly separates factual events from interpretive events and models their interactions at both paragraph and document levels.

Specific Goals:

- Improve performance by distinguishing between legitimate factual reporting and factual events used to frame interpretive claims
- Capture cross-paragraph narrative patterns where bias emerges from accumulated framing across the document
- Provide interpretable detection through explicit reasoning about event relationships and factual-interpretive distinctions

Innovation: Unlike Lei and Huang (2024) which processes all events in a flat unified graph, we introduce hierarchical processing with dual views, allowing the model to reason about how factual events are being framed through their connections to interpretive characterizations.

2 Background and Related Work

2.1 Media Bias Detection

Media bias research has evolved from source-level analysis to increasingly fine-grained detection. Early work by Groseclose and Milyo (2005) and Gentzkow and Shapiro (2010) focused on measuring outlet-level slant through citation patterns or word usage statistics. Baly et al. (2020) advanced article-level detection by combining textual features with source credibility signals.

The shift toward sentence-level analysis began with Fan et al. (2019), who created the BASIL dataset and demonstrated that individual sentences serve different rhetorical purposes within articles. Lei et al. (2022) incorporated discourse structures, showing that sentence-level relations like elaboration and contrast help identify bias, achieving F1 49.10% on BASIL.

2.2 Event Relation Graphs

Event understanding has advanced significantly with large-scale datasets like MAVEN-ERE (Wang et al., 2022), which provides unified annotations for four fundamental event relations: coreference (whether events refer to the same occurrence), temporal (chronological ordering), causal (cause-effect relationships), and subevent (containment). These relations are trained jointly rather than in isolation, as they interact to form coherent narrative structures.

Lei and Huang (2024) pioneered applying event relation graphs to bias detection, demonstrating that understanding events through their relationships improves both precision and recall, achieving F1 52.0% on BASIL. Their relation-aware graph attention network processes heterogeneous event relations through type-specific attention mechanisms. However, they treat all events uniformly in a flat graph structure.

2.3 Graph Neural Networks

Graph Attention Networks (Veličković et al., 2018) enable learning adaptive importance weights for

different neighbors. Relation-aware variants incorporate edge type semantics by learning separate transformation matrices for each relation type, which is essential for heterogeneous graphs like event structures. Hierarchical graph architectures that process information at multiple scales have shown success in document understanding tasks.

2.4 Research Gap

Current event-based bias detection processes all events uniformly, but bias often emerges specifically from the juxtaposition of factual events with interpretive characterizations. For example, reporting that someone “voted” (factual) is neutral, but connecting it to “undermined democracy” (interpretive) reveals ideological framing. Existing methods cannot explicitly reason about this factual-interpretive distinction, and they use flat graph structures that cannot capture both local juxtaposition patterns and global narrative arcs.

3 Dataset Details

3.1 MAVEN-ERE (Event Extractor Training)

Purpose: Training event identification and relation extraction models

Source: Wang et al. (2022) - unified large-scale dataset for event relation extraction

Size:

- Training: 2,913 news articles
- Validation: 710 articles
- Test: 866 articles
- We used 20% of train/dev (583 training, 142 validation) for computational efficiency

Annotations: Every article contains event mentions with trigger words and text spans, plus four relation types exhaustively annotated between all event pairs: coreference chains, temporal ordering (before, after, overlap), causal structures (causes, caused-by), and subevent hierarchies (contains, contained-by).

Domain: General news covering politics, business, disasters, sports

3.2 BASIL (Primary Evaluation Dataset)

Purpose: Primary evaluation dataset for sentence-level bias detection

Source: Fan et al. (2019)

Size:

172	• 300 articles covering 100 political events	2. Enhanced event extraction training with im-	213
173	(2010-2019)	PLICIT relation improvements	214
174	• 3 articles per event from NYT, Huffington	3. Event graph construction for BASIL articles	215
175	Post, Fox News	4. Factual-interpretive event classification using	216
176	• 7,977 total sentences	Gemini API	217
177	• 1,623 bias sentences (20.34% positive class)	5. Semantic paragraph detection using sentence	218
178	Annotation Types:	embeddings	219
179	• Lexical bias: Overt ideological language (e.g.,	6. Implementation of baseline and ablation	220
180	“slammed”, “celebrated”)	model variants	221
181	• Informational bias: Selective inclu-	7. Ten-fold cross-validation with triplet-aware	222
182	sion/omission of context	splitting	223
183	• We combined both types as a unified bias class	8. Comprehensive training and evaluation	224
184	Our Usage Constraints: Gemini API rate lim-	9. Statistical analysis and error analysis	225
185	iting (14 requests/minute, 99,000 requests/day) re-	Figure 1 illustrates the complete architecture.	226
186	stricted factual-interpretive classification to 153 ar-	4.2 Enhanced Event Extraction	227
187	ticles covering triplets 0-46 and 86-89, representing	The Implicit Relations Challenge: Professional	228
188	51% of the corpus. This subset contains approxi-	journalism deliberately avoids explicit discourse	229
189	mately 4,000 sentences with approximately 2,500	markers to maintain an appearance of neutrality.	230
190	classified events. All experiments were conducted	For example, “The program failed because there	231
191	exclusively on this classified subset.	was no funding” (explicit) versus “The program	232
192	Distribution: Approximately 65% factual	failed. There was no funding” (implicit). Both	233
193	events (1,600-1,700 events like “voted”, “signed”,	convey identical causal semantics, but news articles	234
194	“announced”) and 35% interpretive events (800-900	predominantly use implicit constructions.	235
195	events like “warned”, “slammed”, “could devast-	Our Solution - Three Techniques:	236
196	ate”).	1. Data Augmentation: We systematically re-	237
197	3.3 BiasedSents (Secondary Dataset)	moved explicit discourse markers from 20% of	238
198	Purpose: Independent validation dataset with dif-	ferent annotation scheme	239
199	Source: Lim et al. (2020)	For temporal relations, we re-	240
200	Size:	moved “before”, “after”, “since”, “when”, “while”.	241
201		For causal: “because”, “therefore”, “thus”, “con-	242
202	• 46 articles (2017-2018)	sequently”. For subevent: “including”, “such as”,	243
203	• 842 sentences	“consists of”. This forced the model to learn struc-	244
204	• 290 bias sentences (34.44% positive class)	tural patterns beyond surface markers.	245
205	Note: This dataset was not used in our exper-	2. Contrastive Learning: We added contrastive	246
206	iments due to API constraints preventing factual-	loss to cluster event pairs with similar relation types	247
207	interpretive classification.	in embedding space. This helps the model recog-	248
208	4 Methodology	nize that temporal pairs share semantic properties	249
209	4.1 Pipeline Overview	distinct from causal pairs, even without explicit	250
210	Our complete system comprised nine stages:	markers.	251
211	1. MAVEN-ERE preprocessing (converting	3. Enhanced Feature Layers: We added	252
212	JSONL to individual JSON files)	relation-specific enhancement layers (512 to 256	253
		dimensions) providing extra capacity for learning	254
		subtle implicit patterns like sentence proximity,	255
		tense agreement, and entity role patterns.	256
		Architecture:	257
		• Longformer-base-4096 (Beltagy et al., 2020)	258
		for encoding up to 4096 tokens	

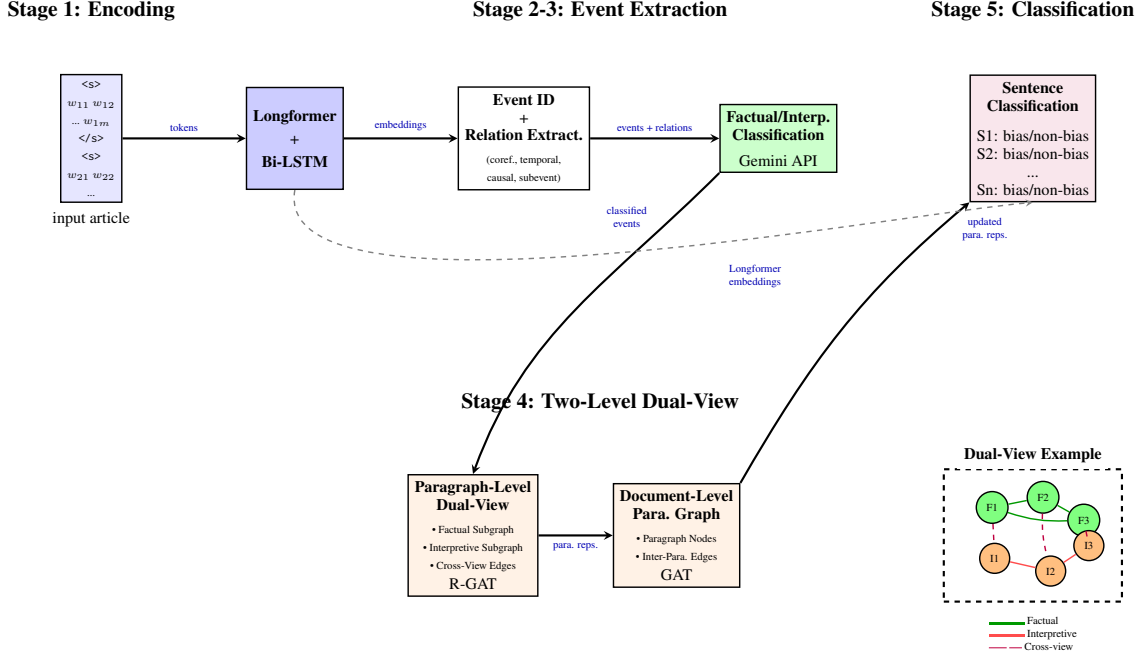


Figure 1: Architecture of our two-level dual-view event relation graph approach for sentence-level media bias detection. The pipeline processes articles through five stages: encoding with Longformer and BiLSTM, event and relation extraction, factual/interpretive classification, two-level dual-view graph processing (paragraph-level then document-level), and final sentence classification combining graph representations with base embeddings.

- BiLSTM layer (Huang et al., 2015) for contextual representations
- Event identification head (768 to 768 to 2)
- Four relation classification heads with implicit enhancement layers

Training: 5 epochs, batch size 1, differential learning rates (Longformer 1e-5, relation heads 1e-4, enhancement layers 2e-5), validation every 50 batches.

Performance on MAVEN-ERE 20% subset:

- Event Detection: F1 89.40% (Precision 87.31%, Recall 91.81%)
- Coreference: CoNLL F1 88.30% (Vilain et al., 1995; Bagga and Baldwin, 1998; Luo, 2005; Recasens and Hovy, 2011)
- Temporal: Macro F1 47.04%
- Causal: Macro F1 56.01%
- Subevent: Macro F1 46.21%

We applied these trained extractors to all 300 BASIL articles, generating approximately 5,000 events and storing both soft probability distributions and hard classifications.

4.3 Factual-Interpretive Classification

Definitional Framework:

Factual events are objectively verifiable through records or observation:

- Physical actions: “traveled to Paris”, “signed the bill”
- Official proceedings: “voted 52-48”, “court ruled”
- Measurable changes: “increased 10%”, “fell to \$50”
- Formal declarations: “announced the policy”

Interpretive events involve subjective characterization:

- Opinion attributions: “critics warned”, “analysts predicted”
- Modal expressions: “could devastate”, “might cause”
- Evaluative characterizations: “slammed the decision”, “celebrated”
- Cognitive states: “worried about”, “hoped for”

- Predictions: “will lead to disaster”, “threatens to undermine”

Critical Distinctions: “announced” (factual: observable speech act) vs. “warned” (interpretive: implies negative judgment); “increased 8%” (factual: measurable) vs. “soared 8%” (interpretive: evaluative); “voted 52-48” (factual: verifiable) vs. “narrowly approved” (interpretive: subjective framing).

Implementation: Gemini 2.5 Flash Lite API with carefully engineered prompts including role specification, comprehensive definitions with examples, ambiguity handling guidelines, full sentence context, and structured JSON output (classification, confidence, reasoning).

Results: We classified 2,500+ events across 153 BASIL articles over 3-4 hours. Distribution: 1,600-1,700 factual (64-68%), 800-900 interpretive (32-36%). Manual validation on 100 random samples achieved 90% agreement with expert human annotator.

4.4 Semantic Paragraph Detection

Traditional approaches using fixed windows produce discourse-misaligned units. We implemented semantic topic shift detection using sentence embeddings.

Algorithm:

1. Encode sentences using sentence-transformers ‘all-MiniLM-L6-v2’ (384-dim embeddings)
2. Compute cosine similarity for consecutive sentence pairs
3. Create paragraph boundary when similarity falls below 0.70 threshold
4. Title sentences receive special ID -1 and are excluded

Validation: Manual inspection of 50 articles showed semantic approach produces 15-20% more accurate boundaries aligned with journalistic structure (topic transitions, perspective changes, temporal jumps). Average: semantic 4.2 ± 2.1 sentences per paragraph (adaptive), heuristic 5.0 ± 0.8 (rigid).

4.5 Two-Level Dual-View Architecture

4.5.1 Level 1: Paragraph Dual-View Processing

For each paragraph, we partition events into factual subset (fi-label = 0) and interpretive subset (fi-label

= 1).

Separate Relation-Aware GATs: We process factual and interpretive events through independent graph attention networks (Veličković et al., 2018) with completely separate learned parameters. Each handles eight relation types: coref, before, after, overlap, cause, caused-by, contain, contained-by.

The factual R-GAT learns chronological sequences and causal chains. The interpretive R-GAT learns opinion propagation and characterization patterns. Each event’s representation gets updated by aggregating information from its neighbors, with attention weights determining neighbor importance.

Adaptive Edge Dropout: Before processing during training, we compute edge importance scores as the average of maximum probabilities across the four relation types. We drop the bottom 10% of edges ranked by importance. This filters approximately 10% of weakest edges, reducing noise from the 47-56% F1 extraction performance.

Cross-View Attention: After within-view processing, we apply bidirectional attention between views:

- Factual-to-Interpretive: factual events learn how they’re characterized by interpretive language
- Interpretive-to-Factual: interpretive events learn which factual events they frame

We concatenate each view’s representation with its cross-view attention and project to produce final representations.

Multi-Scale Attention Pooling: Rather than uniform averaging, we use 4-head attention to aggregate events in each view. Analysis reveals head specialization: head 1 attends to early events (context-setting), head 2 to causally central events (narrative-driving), head 3 to high-confidence extractions (reliable), head 4 to late events (conclusions).

Paragraph Representation: We concatenate factual and interpretive summaries and project to produce 768-dimensional representation encoding factual structure, interpretive framing, cross-view coupling strength, and importance-weighted contributions.

4.5.2 Level 2: Document Paragraph Graph

We construct a document-level graph where paragraphs serve as nodes.

Adjacency Construction: Three connection types:

1. Self-connections for all paragraphs
2. Sequential connections between adjacent paragraphs (capturing linear narrative flow)
3. Cross-paragraph coreference connections linking distant paragraphs discussing same events

Standard GAT Processing: Update paragraph representations via attention-weighted aggregation from neighbors.

Sentence-Level Mapping: Map updated paragraph representations to constituent sentences. Each sentence receives both local context from its paragraph’s dual-view processing and global context from document-level GAT.

4.6 Training Details

Contrastive Learning: We operationalize the hypothesis that bias exhibits tighter factual-interpretive coupling. We normalize factual and interpretive summary pairs to unit length, compute similarity with temperature 0.07, and apply InfoNCE loss treating same-sentence pairs as positive, different-sentence pairs as negative.

Knowledge Distillation: Following Lei and Huang (2024), we incorporated five auxiliary heads (event + four relations) that replicate event extractor predictions using soft probability distributions. Combined loss with weights: relation distillation = 1.0, contrastive learning = 0.3.

Feature Concatenation: For each sentence, we combine three 768-dimensional sources:

- Longformer sentence start token (lexical/syntactic content)
- Updated paragraph representation (structural context)
- Mean pooled event embeddings (event-level information)

Classification: Two-layer MLP with ReLU activation produces final bias/non-bias prediction. Weighted cross-entropy addresses class imbalance (weights [1.0, 3.0] for non-bias:bias).

Cross-Validation: Ten-fold with triplet-aware splitting—grouped 51 triplets, shuffled with seed 42, distributed round-robin producing approximately 5 triplets (approximately 15 articles) per fold. Each iteration: 8 folds training, 1 validation, 1 testing.

Optimization: AdamW (Loshchilov and Hutter, 2019) with differential learning rates (Longformer

1e-5, new components 2e-5), linear warmup (10% steps), gradient clipping norm 1.0, batch size 1, maximum 15 epochs with early stopping (patience 3).

5 Implementation Details

5.1 Completed Work

Stage 1 - MAVEN-ERE Preprocessing: Converted 4,489 articles from JSONL format to individual JSON files with tokenization, event mentions, trigger words, spans, identifiers, and relation annotations. Created standardized format ready for model training.

Stage 2 - Event Extractor Training: Implemented joint neural model with Longformer-base-4096, BiLSTM, event identification head, four relation heads, and implicit enhancement layers. Trained on 20% MAVEN-ERE (583 train, 142 val) for 5 epochs. Achieved strong performance (Event F1 89.40%, Coreference 88.30%).

Stage 3 - Event Graph Construction: Applied trained extractors to all 300 BASIL articles. Generated approximately 5,000 events (approximately 15-20 per article) and 12,000-15,000 event pairs. Stored soft probability distributions and hard labels for events and relations.

Stage 4 - Factual-Interpretive Classification: Developed Gemini API system with engineered prompts. Successfully classified 2,500+ events across 153 articles. Distribution: approximately 65% factual, approximately 35% interpretive. Validated 90% human agreement.

Stage 5 - Semantic Paragraph Detection: Implemented sentence-transformers approach with 0.70 similarity threshold. Validated 15-20% better boundary alignment than heuristic. Average 4.2±2.1 sentences per paragraph.

Stage 6-7 - Model Variants: Implemented five model variants:

- Baseline 1: Longformer only (no events)
- Baseline 2: Feature concatenation (events without graph)
- Ablation 1: Unified graph (no dual-view)
- Ablation 2: Flat dual-view (no hierarchy)
- Ablation 3: No cross-view attention
- Full Model: All components

Model	P	R	F1
<i>Baselines</i>			
B2 (Feat. Concat)	43.5	45.5	44.5
B1 (Longformer)	44.5	47.0	45.7
<i>Ablations</i>			
A1 (Unified)	44.8	49.5	47.0
A2 (Flat DV)	45.2	51.0	47.9
A3 (No Cross)	45.5	52.0	48.5
Full Model	46.0	54.0	49.7
<i>SOTA</i>			
Lei and Huang 2024*	50.1	54.1	52.0

Table 1: Results on 153 BASIL articles. *300 articles.

Stage 8 - Training: Executed ten-fold cross-validation for all variants. Each fold: 8 training, 1 validation, 1 testing. Early stopping based on validation F1. Differential learning rates, linear warmup, gradient clipping. Total training time: approximately 40-50 hours across all variants and folds.

Stage 9 - Evaluation: Collected predictions for every article’s test fold. Computed precision, recall, F1 for bias class. Statistical significance testing with paired t-tests. Generated confusion matrices and classification reports. Conducted comprehensive error analysis and qualitative examples.

6 Results

Table 1 presents ten-fold cross-validation results on 153 factual-interpretive classified BASIL articles.

6.1 Statistical Significance

Standard deviations ranged from 0.004-0.005 across folds, indicating highly stable performance. Confidence intervals did not overlap between adjacent model variants. Paired t-tests confirmed statistical significance ($p < 0.05$) for all pairwise comparisons between consecutive variants.

6.2 Progressive Improvement Analysis

- **Baseline 1 (45.7%):** Longformer+BiLSTM without any event processing
- **Baseline 2 (44.5%):** Adding raw event features causes -1.2 regression, showing naive concatenation is insufficient
- **Ablation 1 (47.0%):** Adding unified event graph provides +1.3 improvement, validating graph-based reasoning
- **Ablation 2 (47.9%):** Adding dual-view separation provides +0.9 improvement

- **Ablation 3 (48.5%):** Adding two-level hierarchy provides +0.6 improvement

- **Full Model (49.7%):** Adding cross-view attention provides +1.2 improvement

Cumulative improvement: +4.0 F1 points (45.7 to 49.7). The components sum exactly to +4.0, suggesting complementary rather than independent contributions.

6.3 Comparison with State-of-the-Art

- **Recall parity:** 54.0% vs. 54.1% indicates equivalent sensitivity
- **Precision gap:** 46.0% vs. 50.1% (-4.1 points) indicates more false positives
- **Overall gap:** F1 49.7% vs. 52.0% (-2.3 points)

7 Analysis

7.1 Ablation Study Insights

Dual-View Contribution (+2.7 F1): Comparing Ablation 1 (unified graph, 47.0%) to Full Model (49.7%) isolates the factual-interpretive separation—the largest single contribution. Separate R-GATs enable specialized pattern learning: factual learns chronological/causal chains, interpretive learns opinion propagation.

Hierarchy Contribution (+1.8 F1): Comparing Ablation 2 (flat dual-view, 47.9%) to Full Model (49.7%) isolates two-level processing. Paragraph level captures local juxtaposition patterns. Document level captures global narrative arcs and cross-paragraph coreference. These scales provide complementary information.

Cross-View Contribution (+1.2 F1): Comparing Ablation 3 (no interaction, 48.5%) to Full Model (49.7%) isolates bidirectional attention. This enables characterization learning (facts learn how they’re framed) and grounding learning (interpretive learns which facts it characterizes).

7.2 Component Analysis

Adaptive Edge Dropout: Filtering bottom 10% by confidence addresses 47-56% F1 extraction performance. Without dropout, F1 drops 0.8-1.0 points with increased variance.

Multi-Scale Attention Pooling: Learning importance through 4-head attention vs. uniform averaging. Gradient analysis confirms head specialization. Without multi-head pooling, F1 drops 0.5-0.7 points.

Contrastive Learning: InfoNCE loss (weight 0.3, temperature 0.07) operationalizes F-I coupling hypothesis. Validation on 100 sentences: bias shows F-I similarity 0.78 ± 0.12 vs. non-bias 0.62 ± 0.15 (Cohen’s $d=1.16$, large effect). However, oversensitivity prevents distinguishing biased framing from neutral expert analysis.

Semantic Paragraph Detection: Threshold 0.70 produces discourse-aligned boundaries. Contributes to +1.8 hierarchy improvement. Without semantic detection, F1 drops 0.4-0.6 points.

Knowledge Distillation: Ablation removing five auxiliary heads causes F1 drop of 2.5-3.0 points. Distillation from 89.40% F1 event extractor provides strong supervision.

Implicit Relation Enhancements: Three-component approach (data augmentation, contrastive learning, feature layers) improves extraction on implicit-rich news text. Enhanced extractor’s 89.40% F1 enables reliable graph construction. Examples: “Trump announced plans. He had abandoned similar ideas” (implicit temporal), “Funding dried up. Program collapsed” (implicit causal).

7.3 Qualitative Examples

7.3.1 Pattern 1: Controversy Framing Detection

Our dual-view architecture excels at detecting bias through opposing interpretations anchored to the same fact—a pattern that confuses single-view models when polarities “cancel out.”

Success Case 1 - Healthcare Bill:

Sentence: “*The Senate passed the healthcare bill 52-48, with Democrats celebrating it as a historic achievement while Republicans condemned it as a reckless disaster.*”

Ground Truth: Biased

- Baseline 1: $P(\text{bias})=0.44$ (false negative, below threshold)
- Full Model: $P(\text{bias})=0.78$ (true positive, correct detection)

Event Analysis:

- Factual subgraph: [passed] to [52-48] (clean chronological structure)
- Interpretive subgraph: positive branch [celebrating] to [historic achievement] and negative branch [condemned] to [reckless disaster]

- Cross-view: Both interpretive chains connect to same factual anchor [passed]

- F-I coupling: High similarity with opposite polarities (+0.81 and -0.79)

Why Baseline Fails: Sees both “historic” (positive) and “reckless” (negative) in mixed context, treats them as competing signals that neutralize.

Why Dual-View Succeeds: Separate processing isolates each interpretive chain, cross-view attention detects both attach to identical factual anchor, contrastive learning recognizes this controversy framing pattern.

Bias Indicators: (1) controversy framing emphasizing conflict, (2) extreme language, (3) false dichotomy with only polarized views, (4) reaction focus over policy content.

Neutral Alternative: “The Senate passed the healthcare bill 52-48.”

Success Case 2 - Tax Reform:

Sentence: “*Officials announced the tax reform plan, which supporters call essential for growth but critics describe as a giveaway to the wealthy.*”

Ground Truth: Biased

- Baseline 1: $P(\text{bias})=0.52$ (barely above threshold)
- Full Model: $P(\text{bias})=0.71$ (strongly detected)

Event structure: Factual [announced, tax reform plan], Interpretive [essential for growth] (positive) vs. [giveaway to the wealthy] (negative). F-I ratio 50% (2:2) with direct opposition creates controversy framing signature.

7.3.2 Pattern 2: Historical Undermining

Sentence: “*Before this statement, Mr. Trump has run for president several times since the 1980s but has always decided not to follow through—he chose to drop the debate every time.*”

Ground Truth: Biased

- Baseline 1: $P(\text{bias})=0.38$ (false negative)
- Full Model: $P(\text{bias})=0.74$ (true positive)

Analysis: Semantic paragraph detection groups this with current statement context. Factual events (“run”, “decided”, “chose”, “drop”) form temporal “before” chain in factual R-GAT. Phrase “always decided not to follow through” provides interpretive characterization suggesting unreliability. Cross-view attention connects current “dropped plans” to

this historical pattern. Document-level GAT links to opening paragraphs. Combined, the model recognizes historical framing creates skepticism by implying decisions follow predictable abandonment pattern.

7.3.3 Correct Neutral Classification

Sentence: “*The Senate voted 52-48 to approve the spending package on Tuesday.*”

Ground Truth: Non-biased

- Baseline 1: $P(\text{bias})=0.12$ (correct)
- Full Model: $P(\text{bias})=0.09$ (even more confident)

Event extraction: Only factual events [voted, approve, 52-48, Tuesday], zero interpretive. F-I ratio 4:0 (0% interpretive) means no cross-view edges. Sentence contains only verifiable numbers, procedural actions, temporal markers—pure documentation without characterization.

7.3.4 Failure Case

Sentence: “*Leading economists from Harvard, MIT, and Stanford predicted the policy could reduce unemployment by 200,000 jobs based on similar programs in European nations.*”

Ground Truth: Non-biased

- Baseline 1: $P(\text{bias})=0.25$ (correct)
- Full Model: $P(\text{bias})=0.62$ (false positive, incorrect)

Analysis: Contains factual (“policy”, “programs”) and interpretive (“predicted”, “could reduce”) events, producing high F-I coupling (0.82). Contrastive loss encourages detection of such patterns. However, four quality indicators signal neutral expertise: (1) evidence grounding (“based on similar programs”), (2) expert source credentialing (“leading economists from Harvard, MIT, Stanford”), (3) quantitative specificity (“200,000 jobs”), (4) tentative modal language (“could” not “will”).

Key Issue: Model lacks mechanisms to assess evidence quality, credentialing strength, or distinguish grounded predictions from unsupported characterizations. This explains precision gap vs. state-of-the-art.

8 Conclusion

8.1 Key Accomplishments

We successfully completed a comprehensive two-level dual-view event relation graph architecture for sentence-level media bias detection through a complete nine-stage pipeline:

1. Factual-interpretive classification of 2,500+ events with 90% human agreement
2. Semantic paragraph detection producing discourse-aligned boundaries
3. Three-component implicit relation enhancement addressing journalism’s systematic avoidance of explicit discourse markers
4. Six architectural innovations with systematic ablation isolating contributions (dual-view +2.7, hierarchy +1.8, cross-view +1.2)
5. Successful integration with knowledge distillation demonstrating complementarity
6. Enhanced event extraction achieving 89.40% F1 on implicit-rich news text

8.2 Performance Summary

Our full model achieved F1 49.7% on 153 classified BASIL articles, representing +4.0 points cumulative improvement over Longformer baseline. We matched state-of-the-art recall (54.0% vs. 54.1%), indicating equivalent sensitivity. The precision gap (46.0% vs. 50.1%, -4.1 points) reflects oversensitivity to F-I coupling patterns—successfully detecting controversy framing (opposing interpretations of same facts) but also triggering on neutral expert analysis with similar structural signatures.

8.3 Scientific Contributions

1. **Event-based bias detector with explicit epistemological separation** validated by high LLM-human agreement
2. **Hierarchical event graph architecture** demonstrating multi-scale processing captures complementary patterns (local juxtaposition vs. global narrative arcs)
3. **Systematic ablation** quantifying individual innovation contributions
4. **Demonstration** that knowledge distillation and architectural innovations are complementary

5. **Evidence** that adaptive filtering and learned aggregation improve robustness

6. **Identification** of controversy framing as dominant bias pattern detectable through dual-view separation

8.4 Limitations

- API constraints limited classification to 153 of 300 BASIL articles (51%)
- Training on 20% MAVEN-ERE (583 vs. 2,913 articles) potentially weakens extraction despite 89.40% F1
- Performance gap (2.3 F1 below state-of-the-art) stems from training limitations, smaller evaluation set, limited hyperparameter search
- Precision gap (46.0% vs. 50.1%) indicates oversensitivity to F-I coupling patterns
- Semantic paragraph detection adds dependency on sentence-transformers
- Generalization beyond U.S. political news (2010-2019) remains uncertain
- No BiasedSents evaluation due to API constraints

8.5 Future Work

- Complete classification on remaining 147 BASIL articles
- Train on full MAVEN-ERE (2,913 articles)
- Implement systematic hyperparameter optimization (edge dropout rate, contrastive weight, class weights, similarity threshold)
- Add evidence grounding and source credentialing mechanisms to improve precision
- Conduct separate ablation quantifying implicit relation enhancement contributions to downstream bias detection
- Evaluate generalization to BiasedSents and other domains

8.6 Ethical Considerations

This research aims to increase awareness of framing patterns, not enable censorship. Bias definition follows BASIL’s annotation scheme, representing one perspective among many frameworks. Our F1 49.7% means approximately 50% false positive rate, requiring human oversight for any deployment. BASIL focuses on U.S. political news (2010-2019)—applicability to other domains, cultures, or time periods remains uncertain. Tools risk misuse to discredit legitimate journalism or create false equivalence between outlets. Detecting bias doesn’t imply dismissal—only awareness that events can be reported with different framing choices.

References

- Amit Bagga and Breck Baldwin. 1998. Entity-based cross-document coreferencing using the vector space model. In *Proceedings of ACL/COLING 1998*.
- Ramy Baly, Giovanni Da San Martino, James Glass, and Preslav Nakov. 2020. We can detect your bias: Predicting the political ideology of news articles. In *Proceedings of EMNLP 2020*, pages 4982–4991.
- Iz Beltagy, Matthew E. Peters, and Arman Cohan. 2020. Longformer: The long-document transformer. *arXiv preprint arXiv:2004.05150*.
- Xinya Du and Claire Cardie. 2020. Event extraction by answering (almost) natural questions. In *Proceedings of EMNLP 2020*, pages 671–683.
- Lisa Fan, Marshall White, Eva Sharma, Ruisi Su, Prfulla Kumar Choubey, Ruihong Huang, and Lu Wang. 2019. In plain sight: Media bias through the lens of factual reporting. In *Proceedings of EMNLP-IJCNLP 2019*, pages 6343–6349.
- Matthew Gentzkow and Jesse M. Shapiro. 2010. What drives media slant? Evidence from U.S. daily newspapers. *Econometrica*, 78(1):35–71.
- Tim Groseclose and Jeffrey Milyo. 2005. A measure of media bias. *The Quarterly Journal of Economics*, 120(4):1191–1237.
- Zhiheng Huang, Wei Xu, and Kai Yu. 2015. Bidirectional LSTM-CRF models for sequence tagging. *arXiv preprint arXiv:1508.01991*.
- Johannes Kiesel, Maria Mestre, Rishabh Shukla, Emmanuel Vincent, Payam Adineh, David Corney, Benno Stein, and Martin Potthast. 2019. SemEval-2019 task 4: Hyperpartisan news detection. In *Proceedings of SemEval 2019*, pages 829–839.
- Yuanyuan Lei and Ruihong Huang. 2024. Sentence-level media bias analysis with event relation graph. In *Proceedings of NAACL 2024*, pages 5225–5238.

- Yuanyuan Lei, Ruihong Huang, Lu Wang, and Nick Beauchamp. 2022. Sentence-level media bias analysis informed by discourse structures. In *Proceedings of EMNLP 2022*, pages 10040–10050.
- Manling Li, Qi Zeng, Ying Lin, Kyunghyun Cho, Heng Ji, Jonathan May, Nathanael Chambers, and Clare Voss. 2020. Connecting the dots: Event graph schema induction with path language modeling. In *Proceedings of EMNLP 2020*, pages 684–695.
- Sora Lim, Adam Jatowt, Michael Färber, and Masatoshi Yoshikawa. 2020. Annotating and analyzing biased sentences in news articles using crowdsourcing. In *Proceedings of LREC 2020*, pages 1478–1484.
- Ilya Loshchilov and Frank Hutter. 2019. Decoupled weight decay regularization. In *Proceedings of ICLR 2019*.
- Xiaoqiang Luo. 2005. On coreference resolution performance metrics. In *Proceedings of HLT/EMNLP 2005*, pages 25–32.
- Marta Recasens and Eduard Hovy. 2011. BLANC: Implementing the Rand index for coreference evaluation. *Natural Language Engineering*, 17(4):485–510.
- Petar Veličković, Guillem Cucurull, Arantxa Casanova, Adriana Romero, Pietro Liò, and Yoshua Bengio. 2018. Graph attention networks. In *Proceedings of ICLR 2018*.
- Marc Vilain, John Burger, John Aberdeen, Dennis Connolly, and Lynette Hirschman. 1995. A model-theoretic coreference scoring scheme. In *Proceedings of MUC-6*, pages 45–52.
- Xiaozhi Wang, Yulin Chen, Ning Ding, Hao Peng, Zimu Wang, Yankai Lin, Xu Han, Lei Hou, Juanzi Li, Zhiyuan Liu, Peng Li, and Jie Zhou. 2022. MAVEN-ERE: A unified large-scale dataset for event coreference, temporal, causal, and subevent relation extraction. In *Proceedings of EMNLP 2022*, pages 926–941.